

CP020003 — Artificial Intelligence 2026

Khon Kaen University

Week 2 — Homework Assignment

1. Overview

This week we practiced feature engineering on a Spotify dataset in class. Now it's your turn to apply the exact same toolkit — pandas transformations, ratio/interaction features, binning, text-based features, and group aggregations — to a brand-new dataset: the famous Netflix Movies and TV Shows catalog.

Good features are the foundation of every strong machine learning model. This assignment has two parts:

- **Part A:** 5 required questions (fixed) — everyone must complete these.
- **Part B:** 2 creative features of your own design — you invent the idea AND the code. Marks depend on creativity and difficulty.

NETFLIX

2. Dataset

Dataset file: [netflix-kku-dataset.csv](#)

Download link: [netflix-kku-dataset.csv \(GitHub\)](#)

Load it directly in pandas using the raw GitHub URL:

```
pd.read_csv("https://raw.githubusercontent.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1/main/dataset/netflix-kku-dataset.csv")
```

2.1 Data Dictionary

Before writing any code, make sure you understand every column below.

Pay special attention to **duration** — it stores different units depending on **type** (minutes for Movies, seasons for TV Shows). This is a very common real-world data-cleaning trap!

Column	Data Type	Description
show_id	string	Unique identifier for each title (e.g. "s1", "s2", ...).
type	categorical	Content type: either "Movie" or "TV Show".
title	string	Name of the movie or TV show.
director	string (may be missing)	Director name(s). Multiple directors are separated by commas. Often missing for TV shows.
cast	string (may be missing)	Main actors, separated by commas. Can contain a large number of names.
country	string (may be missing)	Country (or countries, comma-separated) where the title was produced.
date_added	string / date (may be missing)	The date the title was added to the Netflix catalog, formatted like "September 25, 2021".
release_year	integer	The year the movie or show was originally released.
rating	categorical (may be missing)	Content/age rating such as PG-13, TV-MA, TV-14, etc.
duration	string (mixed units!)	For Movies: length in minutes, e.g. "90 min". For TV Shows: number of seasons, e.g. "2 Seasons". Same column, two different meanings depending on "type" — a classic real-world data cleaning trap!
listed_in	string (multi-label)	One or more genres/categories, separated by commas, e.g. "Crime TV Shows, International TV Shows".
description	string	A short free-text synopsis of the title.

Known data quality notes: director, cast, country, date_added, and rating all contain some missing (NaN) values — always handle these safely (e.g. with .fillna(), .dropna(), or checks like pd.notna()) instead of assuming every cell is filled in.

3. Part A — Required Features (Fixed, 5 Questions)

Create the following 5 new columns in your dataframe. Write clean, commented pandas code for each. These questions are ordered roughly from easier to harder.

Question 1 — Cast Size (★ Easy)

Create a column `cast_size` that counts how many actors are listed in the `cast` column for each title. If `cast` is missing (NaN), the count should be 0.

Hint: Split the string on commas and count the pieces — but check for missing values first.

Question 2 — Genre Count (★ Easy)

Create a column `genre_count` that counts how many genres/categories are listed in the `listed_in` column for each title (e.g. "Crime TV Shows, International TV Shows, TV Action & Adventure" -> 3).

Hint: Same splitting technique as Question 1, applied to a different column.

Question 3 — International Co-Production Flag (★★ Medium)

Create a boolean/int column `is_international` that equals 1 if the `country` column lists more than one country (comma-separated), and 0 otherwise (including when `country` is missing).

Hint: Reuse your comma-splitting logic, then check if the resulting list has more than 1 element.

Question 4 — Duration Parsing by Content Type (★★★★ Medium-Hard)

The `duration` column mixes two different units depending on `type`. Create two new columns: `duration_value` (the numeric part, e.g. 90 or 2) and `duration_unit` (either "min" or "Season(s)"). You will need to use the `type` column to decide how to parse each row correctly.

Hint: Regular expressions (e.g. `re.findall(r'\d+', text)`) or `.str.extract()` can pull the number out. Write a function with `if/elif` logic based on the `type` column, then `.apply()` it row-wise.

Question 5 — Years Until Added to Netflix (★★★★★ Hard)

Create a column `years_until_added` that calculates how many years after its original `release_year` a title was added to the Netflix catalog, using the year portion of `date_added`. Handle missing `date_added` values gracefully (leave the result as NaN rather than crashing).

Hint: You'll need to parse the year out of a string like "September 25, 2021" — try `pd.to_datetime()` on the column, then subtract `.dt.year` from `release_year`.

4. Part B — Design Your Own Creative Features (Bonus, 2 Questions Minimum)

This is the fun part! Invent at least 2 of your own original feature ideas using this dataset. They can combine any columns you like, use text mining on description, build ratios, rankings, group statistics, or anything else creative.

For each idea, write:

- A one-sentence description of what the feature captures and why it might be useful.
- The pandas code that creates it.
- A quick `.head()` or `value_counts()` showing it works.

Grading for Part B is based on: creativity/originality, technical difficulty, and whether the feature is a genuinely useful signal — not just correctness. There is no single "right answer" here.

Need inspiration? Some directions to think about (do NOT just copy these — invent your own!):

- *Combining multiple columns into a single composite "score"*
- *Text mining the description field for themes, keywords, or sentiment*
- *Frequency/count-based features (e.g. how often a director or actor appears)*
- *Domain-knowledge features based on how Netflix or the film industry actually works*

5. Submission Instructions

- Submit a single Jupyter notebook (.ipynb) with all cells run top-to-bottom, no errors.
- Clearly label each question with a markdown header (e.g. "## Question 1").
- Include a short markdown note above each Part B feature explaining your idea.
- Do not modify the original dataset file — always work on a copy or new columns.

6. Grading Rubric (100 points total)

Component	Points	Notes
Part A — 5 Required Features	50 pts (10 each)	Correct logic, correct output, and a short one-line comment explaining what the feature captures.
Part B — 2 Creative Features	40 pts (20 each)	Graded on: (1) creativity/originality, (2) technical difficulty, (3) whether the feature is a genuinely useful signal for future modeling.
Code Quality & Clarity	10 pts	Readable variable names, brief comments, notebook runs top-to-bottom without errors.

7. Dataset Credit

This dataset is a class-hosted copy of the ["Netflix Movies and TV Shows" dataset](#) originally published on **Kaggle** by Shivam Bansal, released under a CC0: Public Domain license. All credit for the original data collection goes to the author and Kaggle.

Good luck, and have fun being creative in Part B! 🍷